

- ① a) Draw a ray diagram to show the working of a compound microscope. Obtain the expression for the total magnification for the final image to be formed at the near point.
 b) In a compound microscope an object is placed at a distance of 1.5 cm from the objective of focal length 1.25 cm . If the eye piece has a focal length of 5 cm , and the final image is formed at near point. Find the magnifying power.
- ② Name two important processes involved in the formation of a p-n junction diode. With the help of a circuit diagram, explain the working of junction diode as a full wave rectifier. Draw its input and output waveforms. State the characteristic property of a junction diode that makes it suitable for rectification.
- ③ State the basic principle behind the working of the ac generator. Briefly describe its working and obtain the expression for the instantaneous value of emf induced.
- ④ The radius of ${}_{13}^{27}\text{X}$ nucleus is R . Then the radius of ${}_{53}^{125}\text{Y}$ will be?
- ⑤ The power of a thin lens is $+5\text{ D}$. When it is immersed in a liquid, it behaves like a concave lens of focal length 100 cm . Calculate the refractive index of the liquid. Given the refractive index of glass is $+1.5$.